

■ 3.8.2 Fourier Transforms

A common operation in analyzing various kinds of data is to find the Fourier transform, or spectrum, of a list of values. The idea is typically to pick out components of the data with particular frequencies, or ranges of frequencies.

<code>Fourier[{y₀, y₁, ...}]</code>	Fourier transform
<code>InverseFourier[{y₀, y₁, ...}]</code>	inverse Fourier transform

Fourier transforms.

Mathematica uses the standard complex Fourier transform. The Fourier transform b_s of a list a_r of length n is defined to be $\frac{1}{\sqrt{n}} \sum_{r=1}^n a_r e^{2\pi i(r-1)(s-1)/n}$. Notice that the zero frequency term appears at position 1 in the resulting list.

The Fourier transform takes you from a time series of data to the frequency components of the data. You can use the inverse Fourier transform to get back to the time series.

The inverse Fourier transform a_r of a list b_s of length n is defined to be $\frac{1}{\sqrt{n}} \sum_{s=1}^n b_s e^{-2\pi i(r-1)(s-1)/n}$.

Here is some data, corresponding to a square pulse.

```
In[1]:= {-1, -1, -1, -1, 1, 1, 1, 1}
Out[1]= {-1, -1, -1, -1, 1, 1, 1, 1}
```

Here is the Fourier transform of the data. It involves complex numbers.

```
In[2]:= fft = Fourier[%]
Out[2]= {0., -0.707107 - 1.70711 I, 0.,
         -0.707107 - 0.292893 I, 0., -0.707107 + 0.292893 I,
         0., -0.707107 + 1.70711 I}
```

Here is the inverse Fourier transform.

```
In[3]:= InverseFourier[%]
Out[3]= {-1., -1., -1. + 2.7477 10-16 I, -1., 1., 1.,
         1. - 2.7477 10-16 I, 1.}
```

After using `Chop`, this gives back exactly your original data.

```
In[4]:= Chop[%]
Out[4]= {-1., -1., -1., -1., 1., 1., 1., 1.}
```

Fourier works whether or not your list of data has a length which is a power of two.

```
In[5]:= Fourier[{1, -1, 1}]
Out[5]= {0.57735, 0.57735 - 1. I, 0.57735 + 1. I}
```

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